Comparison of Signal Quality of Different PPG Sensor Devices Across Different Body Locations and SpO2 Estimation

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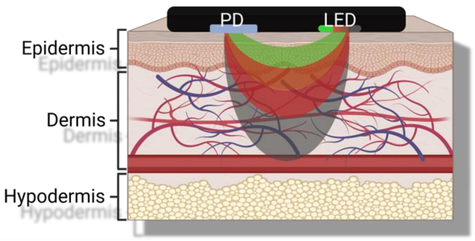
Introduction

PPG Technology: Non-invasive Monitoring of SpO2 and Pulse Rate that uses a light source and photodetector to measure blood volume changes in tissues.

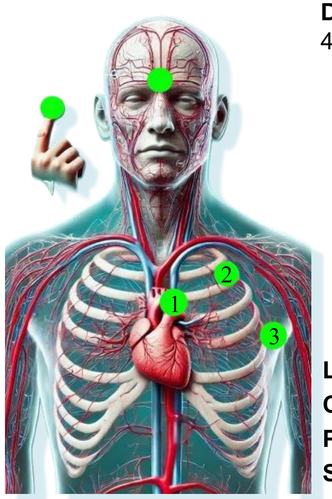
Primary Goal: Compare PPG signal quality using Perfusion Index (PI) and validate SpO2 accuracy.

Perfusion Index (PI):PI indicates PPG signal strength, calculated as the ratio of pulsatile to static blood flow in tissues.

SpO2 Estimation: SpO2 measures the percentage of oxygen-saturated hemoglobin. This study validates SpO2 readings from PPG devices against a MASIMO oximeter.

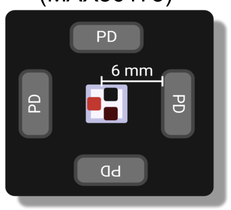


Methods

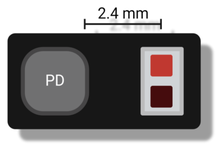


Devices: 2 different devices compared using 4 different healthy subjects

Device 1 (MAX86173)



Device 2 (MAXM86161)



Locations: Chest (3), Forehead, Finger.

Conditions: Pressures (15, 50, 90 mmHg) for chest and forehead.

Protocol: 1 min normal breathing.

SpO2 formula: SpO2 = 110 - 25 *(RED PI / IR PI)

Discussion / Conclusion

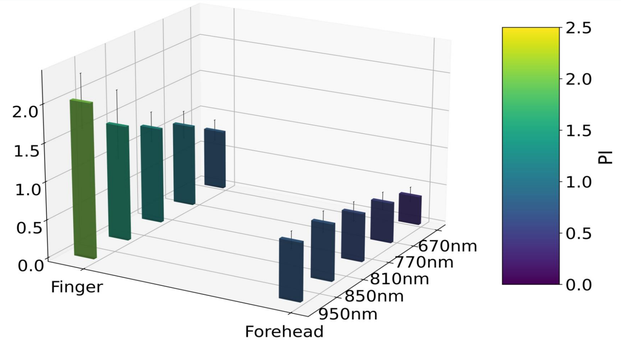
Device comparison: Device 1 exhibits higher PI values due to the increased distance between the LED and photodetector and the use of longer IR wavelengths, enhancing tissue interaction and signal strength.

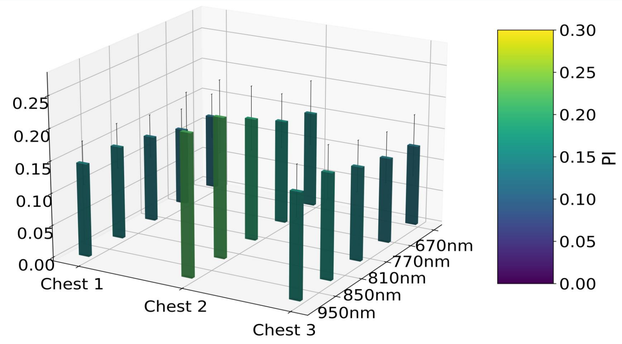
Signal Quality: Varying pressures have minimal impact, with chest location 2 providing the most consistent readings. Finger and forehead measurements also met expectations for high signal quality.

SpO2 Estimation: Using the empirical formula, Device 1's SpO2 estimation shows better statistical outputs, defining the estimation as fairly accurate.

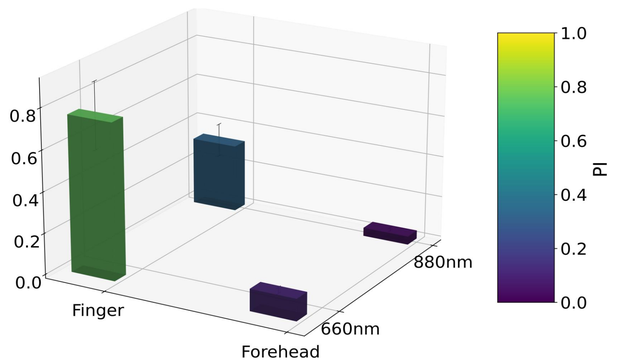
Results

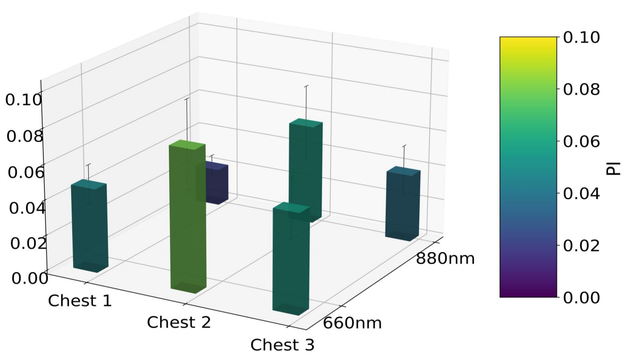
Device 1

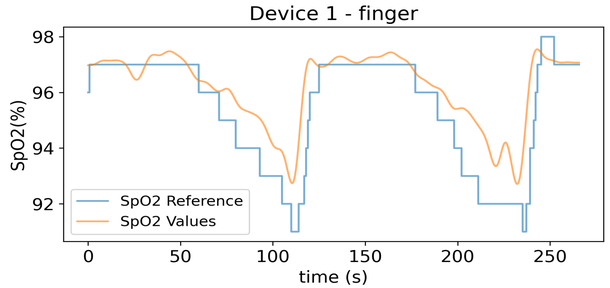




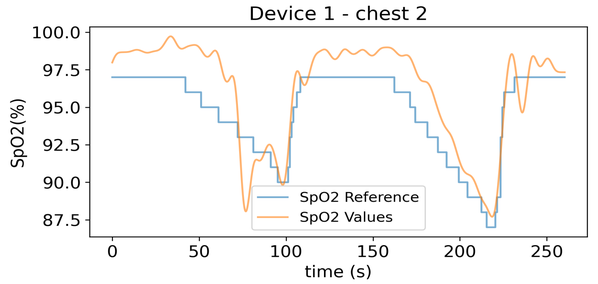
Device 2



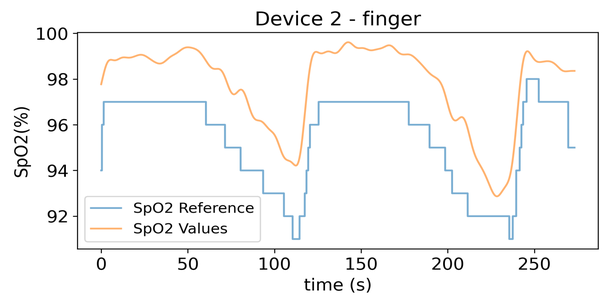




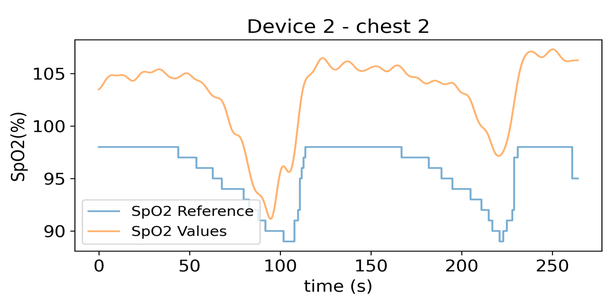
PCC: 0.90 MAE: 0.85 RMSE: 1.25 Within LoA: 95.27%



PCC: 0.75 MAE: 2.16 RMSE: 3.10 Within LoA: 97.87%



PCC: 0.94 MAE: 2.32 RMSE: 2.41 Within LoA: 92.92%



PCC: 0.88 MAE: 7.32 RMSE: 7.57 Within LoA: 92.68%

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